RotationBlind: How Fragile Are Modern Vision Features to Geometric Transformations?

02501 Advanced Deep Learning in Computer Vision

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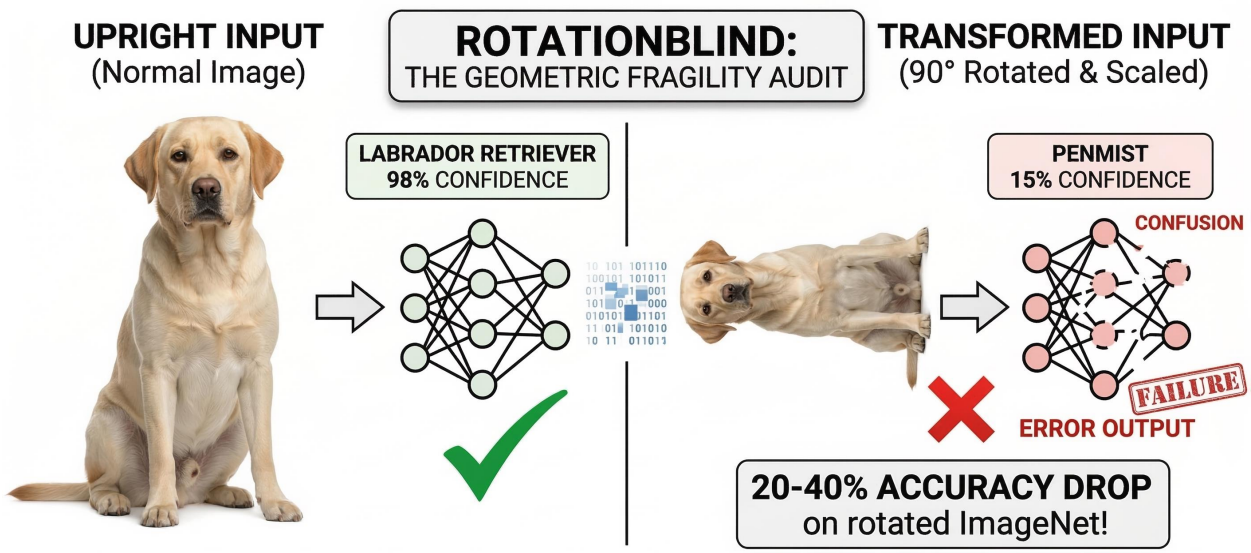

Figure 1: How robust are today's best vision features to rotations, scale changes, and affine transformations?

1 Project description

Classical computer vision placed enormous emphasis on building features that are invariant to geometric transformations: SIFT descriptors are rotation-invariant, HOG features are built from oriented gradients, and early CNNs gained translational equivariance through weight sharing. However, modern vision backbones (ViTs [1], CLIP [2], and DINOv2/v3 [3, 4]) have largely abandoned explicit geometric invariance in favor of learning from massive datasets with data augmentation. This raises a fundamental question: how robust are today's best vision features to rotations, scale changes, and affine transformations?

Recent work has shown that both CNNs and ViTs perform poorly on rotated inputs [5, 6], with accuracy drops of 20-40% on ImageNet for 90-degree rotations. This is alarming because it

means that foundation models powering VLMs, robotics, and medical imaging may silently fail on non-upright images.

The goal of this project is to conduct a comprehensive audit of geometric robustness across modern vision backbones (ViT supervised, CLIP, DINOv2, DINOv3, SigLIP) and then propose lightweight methods to restore geometric invariance without retraining from scratch. This could involve test-time augmentation strategies, learned canonicalization modules [6], or feature-space symmetrization. The project addresses a fundamental gap in our understanding of modern vision features and has clear implications for downstream tasks (detection, segmentation, retrieval) where geometric robustness matters.

2 Data

You can use ImageNet and its rotated variants (ImageNet-R, or self-constructed rotated versions at 0/45/90/135/180/225/270/315 degrees). For dense tasks, use can COCO or ADE20K with geometric augmentations applied at test time, etc, etc.

3 Tasks

In this project, you could work on the following tasks:
Task 1: Comprehensive Geometric Robustness Audit. Systematically evaluate modern vision backbones (ViT-B/L supervised, CLIP ViT-B/L, DINOv2-B/L/G, DINOv3, SigLIP) on classification under rotation (0-360 degrees), scaling (0.5x-2x), and affine transformations. Plot accuracy as a function of transformation parameters. Which backbones are most/least robust? How does model size affect robustness?
Task 2: Feature-Level Analysis. Go beyond accuracy! Analyze how the internal representations change under geometric transformations. Do features rotate equivariantly, or do they break completely? You could use CKA similarity, centered cosine similarity, or representation similarity analysis across layers. Compare supervised, self-supervised, and contrastive models.
Task 3: Lightweight Invariance Restoration. Propose and evaluate methods to improve geometric robustness without full retraining. For example, test-time augmentation with rotation ensembles, or a learned canonicalization module that predicts the "canonical" orientation and rotates back before feature extraction (e.g., [6]), or feature-space averaging over multiple rotated views. Measure accuracy recovery vs. computational overhead.
Task 4: ...
Task 5: ...

References

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